

SQL Analytics Project

End-to-end Data Exploration, Cleaning & Business Insights

PostgreSQL | Data Analytics | E-commerce

Tools: PostgreSQL 16 · pgAdmin · SQL

Project Overview

This project performs a complete SQL-based analysis of Zepto's product catalogue — India's fastest-growing 10-minute grocery delivery platform. Starting from raw data ingestion, the workflow covers schema design, data exploration, thorough cleaning, and eight targeted business queries that generate actionable insights on pricing, inventory, revenue, and value.

Attribute	Detail
Platform	Zepto (Quick-Commerce / Grocery Delivery)
Database	PostgreSQL 16
Table	zepto
Total Columns	10 (sku_id, category, name, mrp, discountPercent, availableQuantity, disc
Primary Key	sku_id (SERIAL)

1 · Database Schema Design

The **zepto** table was designed to capture every SKU (Stock Keeping Unit) listed on the Zepto platform. The schema uses appropriate data types to support numerical analysis on pricing, weight, and inventory.

```
CREATE TABLE zepto (  
  sku_id SERIAL PRIMARY KEY,  
  category VARCHAR(120),  
  name VARCHAR(150) NOT NULL,  
  mrp NUMERIC(8,2),  
  discountPercent NUMERIC(5,2),  
  availableQuantity INTEGER,  
  discountedSellingPrice NUMERIC(8,2),  
  weightInGms INTEGER,  
  outOfStock BOOLEAN,  
  quantity INTEGER  
);
```

Column	Type	Purpose
sku_id	SERIAL PK	Unique identifier per SKU
category	VARCHAR(120)	Product category (e.g., Dairy, Snacks)
name	VARCHAR(150)	Product name
mrp	NUMERIC(8,2)	Maximum Retail Price
discountPercent	NUMERIC(5,2)	Discount offered (%)
availableQuantity	INTEGER	Units currently in stock
discountedSellingPrice	NUMERIC(8,2)	Actual selling price
weightInGms	INTEGER	Product weight in grams
outOfStock	BOOLEAN	Stock availability flag
quantity	INTEGER	Pack quantity

2 · Data Exploration

Before any analysis, the dataset was thoroughly explored to understand its shape, distribution, and any anomalies.

2.1 · Row Count

```
SELECT COUNT(*) FROM zepto;
```

2.2 · NULL Value Audit

All ten columns were checked for NULL values to assess data completeness:

```
SELECT * FROM zepto  
WHERE name IS NULL OR category IS NULL OR mrp IS NULL  
OR discountPercent IS NULL OR discountedSellingPrice IS NULL  
OR weightInGms IS NULL OR availableQuantity IS NULL  
OR outOfStock IS NULL OR quantity IS NULL;
```

2.3 · Category Distribution

```
SELECT DISTINCT category FROM zepto ORDER BY category;
```

2.4 · Stock Status Breakdown

```
SELECT outOfStock, COUNT(sku_id)
FROM zepto
GROUP BY outOfStock;
```

2.5 · Duplicate Product Names

Products can have multiple SKUs (e.g., different pack sizes). This query surfaces names appearing more than once:

```
SELECT name, COUNT(sku_id) AS "Number of SKUs"
FROM zepto
GROUP BY name
HAVING COUNT(sku_id) > 1
ORDER BY COUNT(sku_id) DESC;
```

3 · Data Cleaning

Two critical cleaning steps were performed to ensure accurate downstream analysis.

3.1 · Remove Zero-Price Records

Products with MRP = 0 are data errors (likely test entries). They were identified and permanently deleted:

```
-- Identify
SELECT * FROM zepto WHERE mrp = 0 OR discountedSellingPrice = 0;

-- Remove
DELETE FROM zepto WHERE mrp = 0;
```

3.2 · Currency Conversion — Paise → Rupees

The raw data stored prices in **paise** (1 ₹ = 100 paise). Both price columns were converted to Indian Rupees (₹):

```
UPDATE zepto
SET mrp = mrp / 100.0,
    discountedSellingPrice = discountedSellingPrice / 100.0;

-- Verify
SELECT mrp, discountedSellingPrice FROM zepto LIMIT 5;
```

■ **Impact:** Without this correction, all revenue and pricing metrics would be inflated by 100×, leading to completely misleading business insights.

4 · Business Analysis — Queries & Insights

Eight business questions were formulated and answered using targeted SQL queries. Each query is accompanied by its business rationale.

Q1 · Top 10 Best-Value Products by Discount

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
ORDER BY discountPercent DESC
LIMIT 10;
```

■ **Insight:** Surfaces the highest-discounted products — ideal for promotional campaigns and loss-leader strategy review.

Q2 · High-MRP Products Currently Out of Stock

```
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE AND mrp > 300
ORDER BY mrp DESC;
```

■ **Insight:** Premium products that are out of stock represent immediate revenue loss and customer dissatisfaction risk.

Q3 · Estimated Revenue by Category

```
SELECT category,
       SUM(discountedSellingPrice * availableQuantity) AS total_revenue
FROM zepto
GROUP BY category
ORDER BY total_revenue DESC;
```

■ **Insight:** Reveals which categories drive the most potential revenue — critical for inventory prioritisation and marketing budget allocation.

Q4 · Premium Products with Low Discounts

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
WHERE mrp > 500 AND discountPercent < 10
ORDER BY mrp DESC, discountPercent DESC;
```

■ **Insight:** Identifies premium items with little discount headroom — these may need competitive repricing or bundling strategies.

Q5 · Top 5 Categories by Average Discount

```
SELECT category,
       ROUND(AVG(discountPercent), 2) AS avg_discount
FROM zepto
GROUP BY category
ORDER BY avg_discount DESC
LIMIT 5;
```

■ **Insight:** Helps buyers understand which categories are most aggressively discounted, potentially signalling margin pressure.

Q6 · Best Price-Per-Gram Value (Products ≥ 100g)

```
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
       ROUND(discountedSellingPrice / weightInGms, 2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram ASC;
```

■ **Insight:** Empowers value-conscious customers and enables Zepto to highlight 'best bang for buck' products in their app.

Q7 · Weight-Based Product Segmentation

```
SELECT DISTINCT name, weightInGms,
       CASE WHEN weightInGms < 1000 THEN 'Low'
            WHEN weightInGms < 5000 THEN 'Medium'
            ELSE 'Bulk'
```

```
END AS weight_category
```

```
FROM zepto;
```

■ **Insight:** Categorises SKUs for logistics planning — bulk items need different handling, storage, and delivery slot allocation.

Q8 · Total Inventory Weight Per Category

```
SELECT category,
```

```
       SUM(weightInGms * availableQuantity) AS total_weight
```

```
FROM zepto
```

```
GROUP BY category
```

```
ORDER BY total_weight DESC;
```

■ **Insight:** Critical for warehouse capacity planning — knowing the physical weight of inventory per category guides shelving and cold-storage allocation.

5 · Key Takeaways

Area	Finding
Data Quality	Zero-price records removed; paise→rupee conversion critical for accuracy
Inventory Risk	Several premium (■300+) products are out of stock, risking revenue leakage
Revenue Focus	Category-level revenue aggregation enables targeted restocking decisions
Pricing Strategy	Some high-MRP products carry <10% discount — room for competitive
Logistics	Weight segmentation (Low/Medium/Bulk) supports efficient last-mile delivery
Customer Value	Price-per-gram metric helps surface genuinely good deals for value shoppers
Promotions	Top-discounted products identified for featured placements or flash sales
Warehouse	Total inventory weight per category guides cold-chain and storage planning

6 · Tech Stack

Component	Detail
Database	PostgreSQL 16
IDE	pgAdmin 4
Language	SQL (DDL + DML + DQL)
Concepts Used	SERIAL PK, NUMERIC types, BOOLEAN flags, GROUP BY, HAVING, C
Domain	E-commerce / Quick-Commerce Analytics